Electro-thermo-mechanical coupling in functionally graded composite laminates: A Displacement-Function approach

Abstract

Reliable and efficient modeling of the coupled electro-thermo-mechanical response of functionally graded metal-matrix composite laminates is of utmost practical importance for the safe and economic design of multifunctional load-bearing structures subjected to electrical and thermal environments. In the present research, a novel displacement-function-based semi-analytical framework has been developed for investigating the multiphysics behavior of functionally graded composite laminates under current-induced electro-thermal loading. The electrical, thermal, and mechanical fields are sequentially coupled by incorporating temperature-dependent electro-thermal material properties and Joule heating effects. In the proposed formulation, the coupled thermomechanical equilibrium problem is modeled through a single scalar displacement function, governed by a fourth-order partial differential equation, from which laminate strains and ply-level stresses are systematically derived. The proposed modeling scheme is applied to Al-SiC functionally graded symmetric laminates with gradually varying SiC reinforcement volume fraction across the thickness, characterized by a power-law grading function. Parametric investigations reveal that the grading profile, applied current density, and internal heat generation strongly govern the electrical potential distribution, temperature evolution, and stress development within the laminate. The predicted electro-thermo-mechanical responses are validated against finite-element simulations, showing excellent agreement and confirming the accuracy of the model. The proposed formulation provides an efficient and physically transparent framework for analyzing electro-thermo-mechanical interactions in functionally graded composite laminates and offers a viable alternative to computationally intensive conventional numerical approaches for advanced multifunctional material design.

Keywords: Functionally graded laminates; Metal-matrix composites; Multiphysics modeling; Thermal stress; Electro-thermo-mechanical coupling; Displacement function formulation

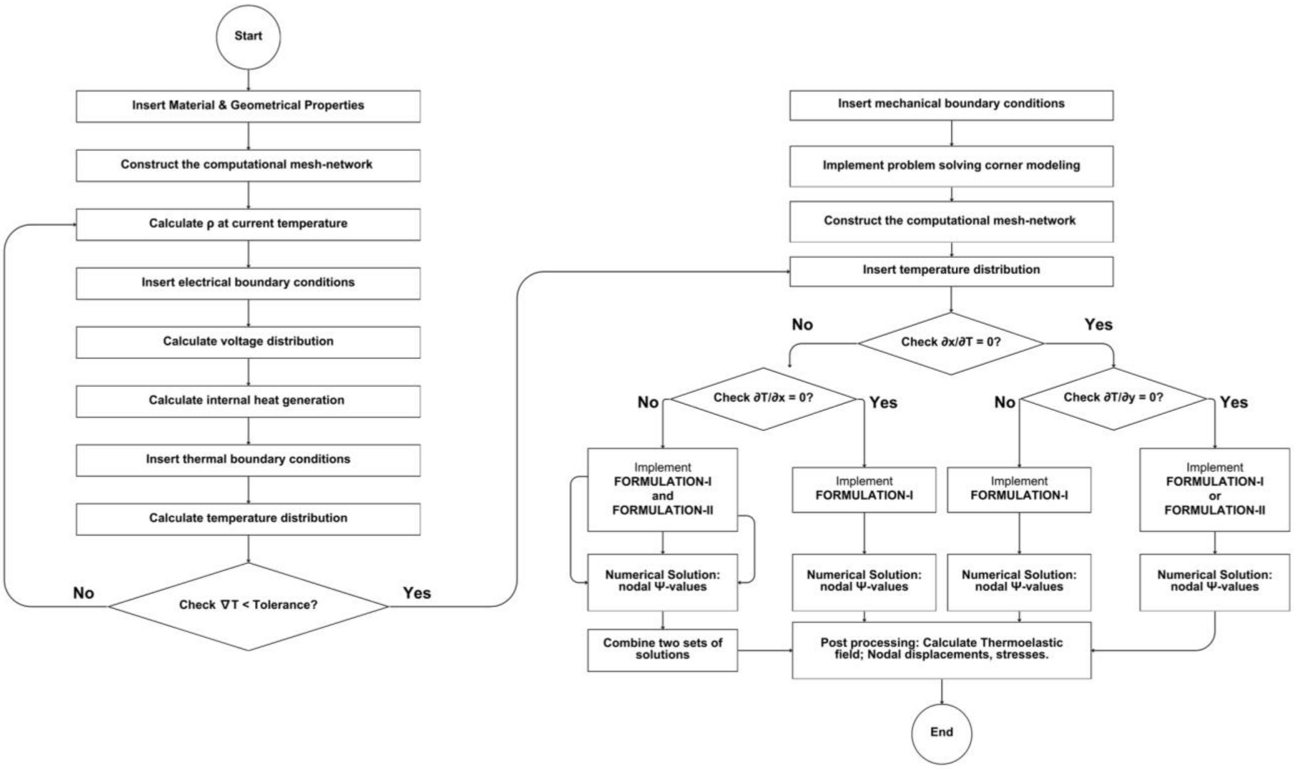

Figure A.1: Flow Chart of the study

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